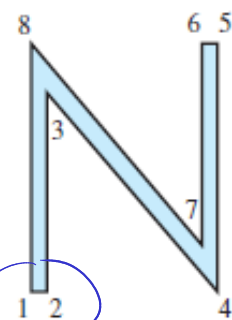


## 2.7 APPLICATIONS TO COMPUTER GRAPHICS

**EXAMPLE 1** The capital letter N in Figure 1 is determined by eight points, or *vertices*. The coordinates of the points can be stored in a data matrix,  $D$ .

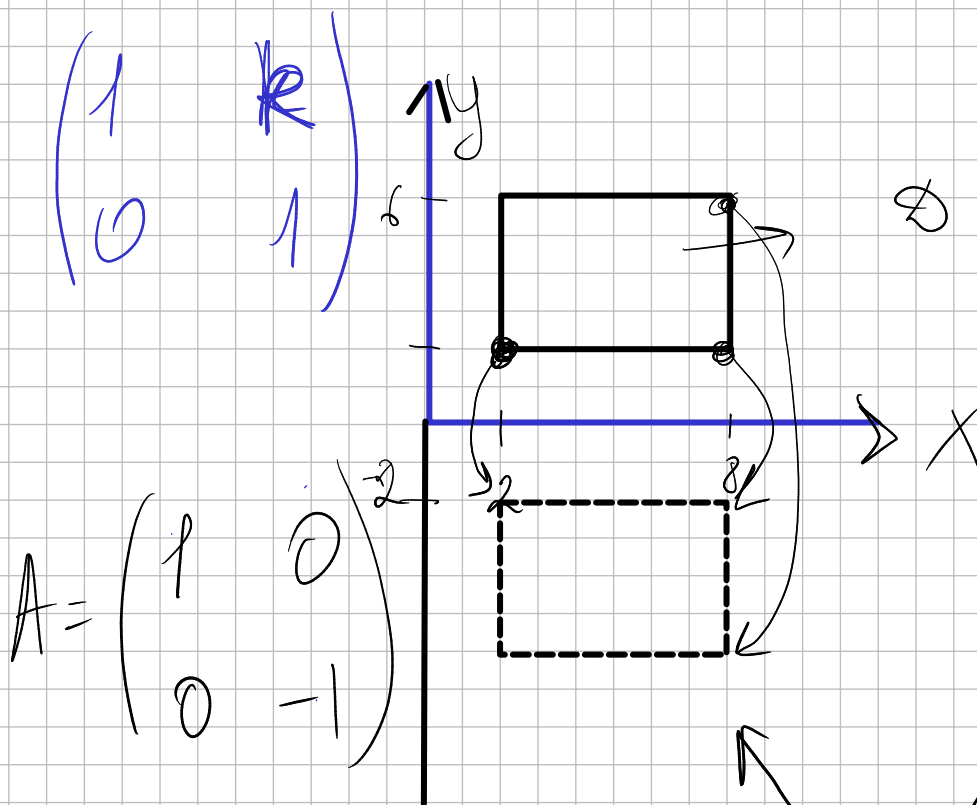
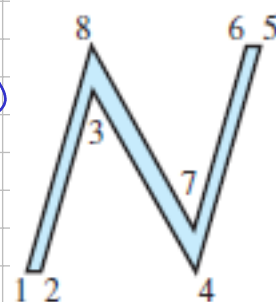
$$\begin{array}{c} \text{Vertex:} \\ \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8 \\ \text{x-coordinate} \quad \left[ \begin{array}{cccccc} 0 & .5 & .5 & 6 & 6 & 5.5 & 5.5 & 0 \end{array} \right] \\ \text{y-coordinate} \quad \left[ \begin{array}{cccccc} 0 & 0 & 6.42 & 0 & 8 & 8 & 1.58 & 8 \end{array} \right] \end{array} = D \end{array}$$

In addition to  $D$ , it is necessary to specify which vertices are connected by lines, but we omit this detail. ■



**EXAMPLE 2** Given  $A = \begin{bmatrix} 1 & .25 \\ 0 & 1 \end{bmatrix}$ , describe the effect of the shear transformation  $\mathbf{x} \mapsto A\mathbf{x}$  on the letter N in Example 1.

$$AD = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 0 & .5 & 2.105 & 6 & 8 & 7.5 & 5.895 & 2 \\ 0 & 0 & 6.420 & 0 & 8 & 8 & 1.580 & 8 \end{bmatrix}$$



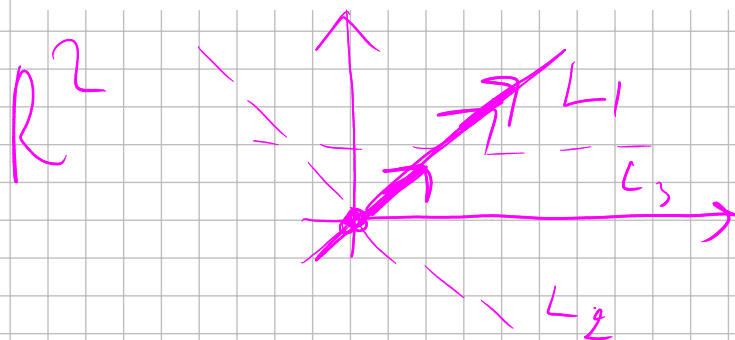
$$D = \begin{pmatrix} 2 & 8 & 8 & 2 \\ 2 & 2 & 6 & 6 \end{pmatrix}$$

$$AD = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 8 & 8 & 2 \\ 2 & 2 & 6 & 6 \end{pmatrix} = \begin{pmatrix} 2 & 8 & 8 & 2 \\ -2 & -2 & -6 & -6 \end{pmatrix}$$

## 2.8 SUBSPACES OF $\mathbb{R}^n$

A subspace of  $\mathbb{R}^n$  is any set  $H$  in  $\mathbb{R}^n$  that has three properties:

- The zero vector is in  $H$ .
- For each  $\mathbf{u}$  and  $\mathbf{v}$  in  $H$ , the sum  $\mathbf{u} + \mathbf{v}$  is in  $H$ .
- For each  $\mathbf{u}$  in  $H$  and each scalar  $c$ , the vector  $c\mathbf{u}$  is in  $H$ .



$L_1$  is a subspace of  $\mathbb{R}^2$   
Also  $L_2$ , but not  $L_3$

### Column Space and Null Space of a Matrix

Subspaces of  $\mathbb{R}^n$  usually occur in applications and theory in one of two ways. In both cases, the subspace can be related to a matrix.

The **column space** of a matrix  $A$  is the set Col  $A$  of all linear combinations of the columns of  $A$ .

If  $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ , with the columns in  $\mathbb{R}^m$ , then  $\text{Col } A$  is the same as  $\text{Span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ . Example 4 shows that the **column space of an  $m \times n$  matrix is a subspace of  $\mathbb{R}^m$** . Note that  $\text{Col } A$  equals  $\mathbb{R}^m$  only when the columns of  $A$  span  $\mathbb{R}^m$ . Otherwise,  $\text{Col } A$  is only part of  $\mathbb{R}^m$ .

**EXAMPLE 4** Let  $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$  and  $\mathbf{b} = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$ . Determine whether  $\mathbf{b}$  is in the column space of  $A$ .

$$c_1 \begin{bmatrix} 1 \\ -4 \\ -3 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 6 \\ 7 \end{bmatrix} + c_3 \begin{bmatrix} -4 \\ -2 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$$

$$\begin{pmatrix} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -10 & 15 \\ 0 & -2 & -6 & 5 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & -4 & 3 \\ 0 & -2 & -6 & 5 \\ 0 & 0 & -2 & 0 \end{pmatrix}$$

$L$ . Row  
 $\begin{bmatrix} 0 & 0 & 1 & b \end{bmatrix}$   
 $\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$

$$\begin{cases} c_1 - 3c_2 - 4c_3 = 3 \\ -2c_2 - 6c_3 = 5 \\ -2c_3 = 0 \end{cases} \Rightarrow \begin{cases} c_3 = 0 \\ c_2 = -2.5 \\ c_1 = -4.5 \end{cases}$$

Yes,  $\mathbf{b} \in \text{Col } A$

The **null space** of a matrix  $A$  is the set  $\text{Nul } A$  of all solutions of the homogeneous equation  $A\mathbf{x} = \mathbf{0}$ .

When  $A$  has  $n$  columns, the solutions of  $A\mathbf{x} = \mathbf{0}$  belong to  $\mathbb{R}^n$ , and the null space of  $A$  is a subset of  $\mathbb{R}^n$ . In fact,  $\text{Nul } A$  has the properties of a *subspace* of  $\mathbb{R}^n$ .

The null space of an  $m \times n$  matrix  $A$  is a subspace of  $\mathbb{R}^n$ . Equivalently, the set of all solutions of a system  $A\mathbf{x} = \mathbf{0}$  of  $m$  homogeneous linear equations in  $n$  unknowns is a subspace of  $\mathbb{R}^n$ .

## Basis for a Subspace

Because a subspace typically contains an infinite number of vectors, some problems involving a subspace are handled best by working with a small finite set of vectors that span the subspace. The smaller the set, the better. It can be shown that the smallest possible spanning set must be linearly independent.

A **basis** for a subspace  $H$  of  $\mathbb{R}^n$  is a linearly independent set in  $H$  that spans  $H$ .

**EXAMPLE 5** The columns of an invertible  $n \times n$  matrix form a basis for all of  $\mathbb{R}^n$  because they are linearly independent and span  $\mathbb{R}^n$ , by the Invertible Matrix Theorem. One such matrix is the  $n \times n$  identity matrix. Its columns are denoted by  $\mathbf{e}_1, \dots, \mathbf{e}_n$ :

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

The set  $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$  is called the **standard basis** for  $\mathbb{R}^n$ . See Figure 3. ■

$R_2 \rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$  — generate the whole  $R_2$  — Span

$\begin{bmatrix} 3 \\ 5 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$R_2 = \text{Span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$

**EXAMPLE 6** Find a basis for the null space of the matrix

$$A = \begin{bmatrix} -3x_1 & 6x_2 & -1x_3 & 1x_4 & -7x_5 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \begin{matrix} = 0 \\ = 0 \\ = 0 \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \\ -3 & 6 & -1 & 1 & -7 \end{bmatrix} \begin{matrix} 0 \\ 0 \\ 0 \end{matrix} \sim \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 5 & 10 & -10 \end{bmatrix} \begin{matrix} R_1 - 2 \cdot R_2 \\ R_3 - 5 \cdot R_2 \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} x_1, x_3 - \text{basic} \\ x_2, x_4, x_5 - \text{free} \end{matrix}$$

$$\begin{cases} x_1 - 2x_2 - x_4 - 3x_5 = 0 \\ x_3 + 2x_4 - 2x_5 = 0 \end{cases}$$

$$x_1 = 2x_2 + x_4 - 3x_5$$

$$x_3 = -2x_4 + 2x_5$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2s + t - 3u \\ s \\ -2t + 2u \\ t \\ u \end{bmatrix} = s \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + u \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

par. solution  
of hom. eqs.

$$\text{Nul } A = \text{Span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

inf many

$$\text{Basis} = \{v_1, v_2, v_3\}$$

only 3

**EXAMPLE 7** Find a basis for the column space of the matrix

$$B = \begin{bmatrix} 1 & 0 & -3 & 5 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{array}{ccccc} & \uparrow & \uparrow & & \uparrow \\ & c_1 & c_2 & & c_5 \end{array}$$

Basis =  $\{c_1, c_2, c_5\}$  / 3 vectors

$$\text{Col } A = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\} \text{ inf many vectors}$$

$$c_3 = -3 \cdot c_1 + 2c_2 = -3 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

## 2.9 DIMENSION AND RANK

### The Dimension of a Subspace

It can be shown that if a subspace  $H$  has a basis of  $p$  vectors, then every basis of  $H$  must consist of exactly  $p$  vectors. (See Exercises 27 and 28.) Thus the following definition makes sense.

The **dimension** of a nonzero subspace  $H$ , denoted by  $\dim H$ , is the number of vectors in any basis for  $H$ . The dimension of the zero subspace  $\{0\}$  is defined to be zero.<sup>2</sup>

The space  $\mathbb{R}^n$  has dimension  $n$ . Every basis for  $\mathbb{R}^n$  consists of  $n$  vectors. A plane through  $0$  in  $\mathbb{R}^3$  is two-dimensional, and a line through  $0$  is one-dimensional.

**EXAMPLE 2** Recall that the null space of the matrix  $A$  in Example 6 in Section 2.8 had a basis of 3 vectors. So the dimension of  $\text{Nul } A$  in this case is 3. Observe how each basis vector corresponds to a free variable in the equation  $Ax = 0$ . Our construction always produces a basis in this way. So, to find the dimension of  $\text{Nul } A$ , simply identify and count the number of free variables in  $Ax = 0$ . ■

The **rank** of a matrix  $A$ , denoted by  $\text{rank } A$ , is the dimension of the column space of  $A$ .

**EXAMPLE 3** Determine the rank of the matrix

$$A = \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 4 & 7 & -4 & -3 & 9 \\ 6 & 9 & -5 & 2 & 4 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & -6 & 4 & 14 & -20 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix} \quad \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 4 & -6 \\ 0 & 0 & 0 & -10 & 15 \end{bmatrix} \quad \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 2 & -3 \\ 0 & 0 & 0 & 2 & -3 \end{bmatrix}$$
$$\begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 2 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{rk } A = 3$$

# Rank and the Invertible Matrix Theorem

The various vector space concepts associated with a matrix provide several more statements for the Invertible Matrix Theorem. They are presented below to follow the statements in the original theorem in Section 2.3.

## The Invertible Matrix Theorem (continued)

Let  $A$  be an  $n \times n$  matrix. Then the following statements are each equivalent to the statement that  $A$  is an invertible matrix.

m. The columns of  $A$  form a basis of  $\mathbb{R}^n$ .

n.  $\text{Col } A = \mathbb{R}^n$

o.  $\dim \text{Col } A = n$

p.  $\text{rank } A = n$

q.  $\text{Nul } A = \{0\}$

r.  $\dim \text{Nul } A = 0$